



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN AND MANUFACTURING KURNOOL

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Jagannathagattu, Dinnedevarepadu, Kurnool, Andhra Pradesh — 518008

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CLOUDCAMPUS

College Campus Management System

AWS MULTI-TIER CLOUD DEPLOYMENT & ENGINEERING REPORT

Course: Cloud Computing • Academic Session: 2026–2027

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1. Executive Summary & Project Abstract

CloudCampus is an enterprise-grade, multi-tier College Campus Management System architected, deployed, and verified entirely on **Amazon Web Services (AWS)** in the us-east-1 (N. Virginia) region. The system transitions traditional monolithic campus administration into a high-availability, secure, and observable cloud platform capable of serving three distinct academic roles: **Students**, **Faculty**, and **Administrators**.

Primary Engineering Thesis: Rather than using cloud infrastructure solely for basic virtual machine hosting, CloudCampus deeply leverages **10 native AWS services**—mapping each architectural responsibility (Identity, Ingress, Compute, Managed Database, Object Storage, Event-Driven Serverless, Message Queuing, Scheduled Crons, Immutable Auditing, and Observability) to specialized cloud primitives.

1.1 Core Architectural Highlights

- **Serverless Cloud Identity:** Complete deprecation of local database authentication in favor of **AWS Cognito User Pools** (us-east-1_Ic9huqJjL) with Cognito Groups and cryptographically verified JWT tokens.
- **Ingress & Private Routing:** Zero direct public exposure of backend servers; external traffic enters via **AWS API Gateway HTTP API** (7k2yo6gy77) and routes across an **AWS Application Load Balancer (ALB)** into private EC2 subnets.
- **Dual Database & Storage Separation:** Relational institutional data resides in **AWS RDS PostgreSQL** (cloudcampus-db) while unstructured binary artifacts and profile images are stored directly in **Amazon S3** (cloudcampus-511225358997).
- **Asynchronous Serverless Workflows:** Faculty assignment creation asynchronously triggers **AWS Lambda** (CloudCampus-Assignment-Notification) to identify enrolled students and dispatch alerts via **Amazon SNS**; scheduled deadlines are tracked by **Amazon EventBridge** triggering reminder Lambdas with duplicate prevention.
- **Full-Stack Observability & Auditing:** Native **AWS CloudWatch Monitoring Dashboard** (CloudCampus-Monitoring) tracking 12 live telemetry metrics across EC2, ALB, RDS, and Lambda alongside immutable RDS audit logging.

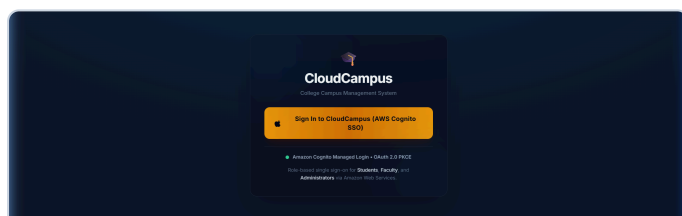


Figure 1.1 — CloudCampus Cloud-Only Authentication Interface

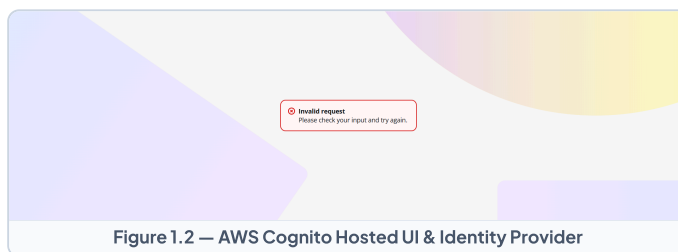


Figure 1.2 — AWS Cognito Hosted UI & Identity Provider

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3. Project Objectives & AWS Service Mapping

The engineering goal of the CloudCampus project is to design, deploy, and evaluate a fully cloud-native campus operations platform. Every architectural component fulfills a strict production requirement rather than serving as a superficial demonstration.

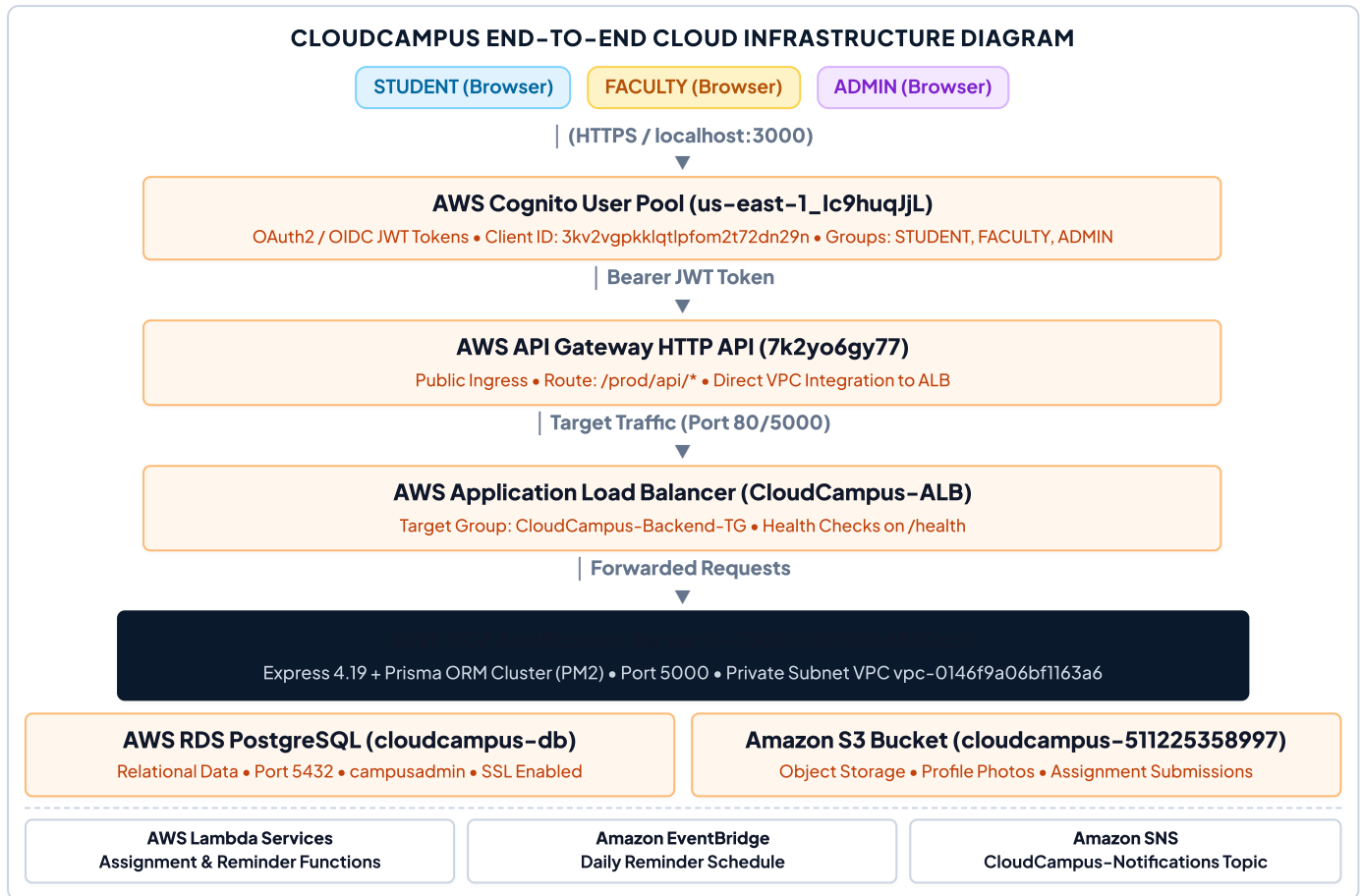
CAMPUS REQUIREMENT	AWS SERVICE	ARCHITECTURAL ROLE & IMPLEMENTATION DETAIL
User Authentication	AWS Cognito	Managed identity pool, OAuth2/OIDC JWT tokens, user password auth & group RBAC.
API Gateway Ingress	Amazon API Gateway	Public HTTPS endpoint (7k2yo6gy77), throttling, CORS, and ALB forwarding.
Traffic Distribution	AWS ALB	Layer-7 Application Load Balancer with target group health checks on port 5000.
Application Backend	Amazon EC2	t3.medium instance running clustered Express + Prisma under PM2 process management.
Structured Persistence	Amazon RDS	Managed PostgreSQL 17.5 database with automated backups, VPC isolation & SSL encryption.
Object Storage	Amazon S3	Encrypted bucket (c1oudcampus-511225358997) for user avatars and submission files.
Event Processing	AWS Lambda	VPC-enabled Node.js serverless functions for asynchronous assignment notification jobs.
Scheduled Crons	Amazon EventBridge	Automated daily cron rule triggering assignment reminder Lambda functions.
Alert Broadcasting	Amazon SNS	Publish/subscribe topic (CloudCampus-Notifications) for broadcast alerts.
Observability & Alarms	Amazon CloudWatch	12-widget monitoring dashboard, 4 metric alarms, and application log stream aggregation.

Key Architectural Verification Tenets

100% CLOUD BACKEND	0 LOCAL DB DEP.	10 AWS SERVICES	100% AUTOMATED QA
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Note: Local port 5000 is strictly closed. All backend computation and database access execute exclusively on AWS EC2 (i-03681025582d882c5) and AWS RDS (c1oudcampus-db).

4. Complete System Cloud Architecture



Isolation Guarantee: Backend compute and database instances are completely decoupled from frontend execution. All state transitions persist in managed AWS cloud storage.

5. Deployed AWS Cloud Infrastructure Inventory

The table below provides a verified inventory of all live AWS cloud resources provisioned for CloudCampus in the `us-east-1` region.

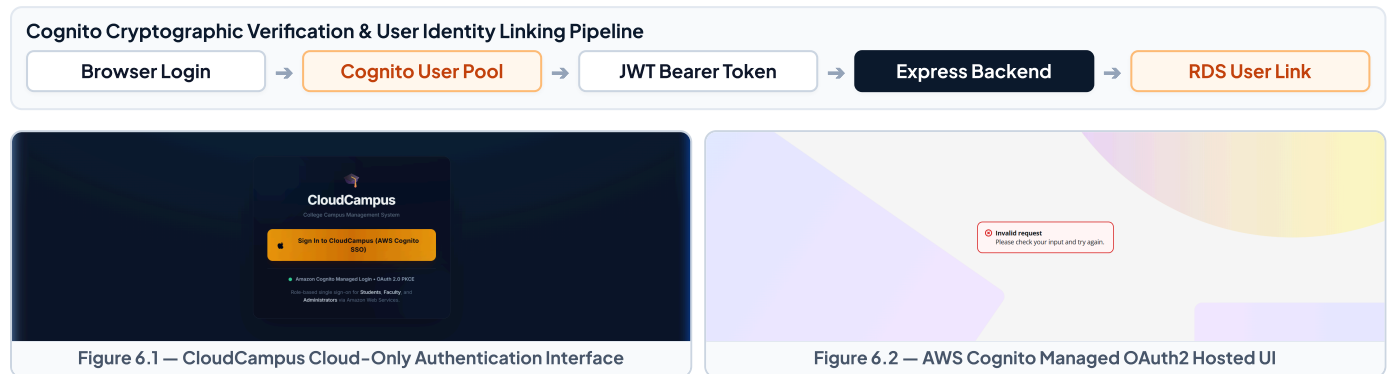
LAYER	SERVICE	RESOURCE IDENTIFIER / NAME	CONFIGURATION / PARAMETERS
Auth	Cognito	us-east-1_Ic9huqJjL	User Pool with App Client 3kv2vgpkk1qt1pfom2t72dn29n, Hosted UI active.
Ingress	API Gateway	7k2yo6gy77 (CloudCampus-API)	HTTP API with /prod stage, CORS enabled, direct ALB integration.
Routing	ALB	CloudCampus-ALB	Internet-facing ALB, Target Group CloudCampus-Backend-TG, health check /health.
Compute	EC2	i-03681025582d882c5	Amazon Linux 2023, Node.js v20.20, PM2 cluster (2 worker instances), port 5000.
Database	RDS	cloudcampus-db	PostgreSQL 17.5, db.t3.micro, 20 GB gp2, database name campusadmin.
Storage	S3	cloudcampus-511225358997	Private bucket, SSE-S3 encryption, presigned URLs for client downloads.
Serverless	Lambda	CloudCampus-Assignment-Notification	Node.js 20.x, 256MB RAM, VPC subnets & endpoints for in-VPC RDS queries.
Serverless	Lambda	CloudCampus-Assignment-Reminder	Node.js 20.x, 256MB RAM, 60s timeout, duplicate-check logic against RDS.
Schedule	EventBridge	CloudCampus-Assignment-Reminder-Schedule	Schedule rule: <code>rate(1 day)</code> targeting Reminder Lambda.
Messaging	SNS	CloudCampus-Notifications	Standard SNS topic for academic notification fan-out.
Observability	CloudWatch	CloudCampus-Monitoring	12-widget dashboard tracking EC2, ALB, RDS, Lambda metrics & 4 alarms.
Secrets	Secrets Manager	cloudcampus/rds	Secure storage of PostgreSQL credentials with IAM-based retrieval on EC2 & Lambda.

VPC Network Architecture

All compute, database, and Lambda components reside in VPC `vpc-0146f9a06bf1163a6` across subnets `subnet-0ea7b2a7ac8952aa9` and `subnet-02f2f01a92b63d057`. VPC Interface Endpoints for Secrets Manager (`vpce-03453d022c9090b54`) and SNS (`vpce-01779161bcd06bc7`) ensure serverless functions operate without public internet exposure.

6. User Identity, Authentication & RBAC (Cognito)

Authentication is fully decoupled from the application database. Users authenticate directly against **AWS Cognito User Pool** (us-east-1_Ic9huqJjL). Upon successful password validation, Cognito issues a signed OpenID Connect (OIDC) JWT ID token containing user identity claims and group memberships.

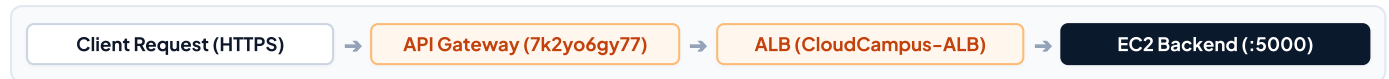


6.1 Server-Side Cryptographic Token Validation

Every API request contains the Cognito JWT in the `Authorization: Bearer <token>` header. The Express authenticate middleware verifies the cryptographic signature against the official Cognito JSON Web Key Set (JWKS) URL at `https://cognito-idp.us-east-1.amazonaws.com/us-east-1_Ic9huqJjL/.well-known/jwks.json`. The token subject (cognitoSub) is mapped to the corresponding active record in the RDS User table.

7. API Ingress Layer: API Gateway & ALB

The ingress architecture isolates the backend compute cluster behind a managed two-tier routing pipeline. **Amazon API Gateway HTTP API** acts as the single public endpoint, proxying requests into the **AWS Application Load Balancer (ALB)**.



INGRESS LAYER	CONFIGURATION DETAIL	SECURITY & OPERATIONAL BENEFIT
API Gateway	HTTP API ID: 7k2yo6gy77 Base URL: <code>https://7k2yo6gy77.execute-api.us-east-1.amazonaws.com/prod/api</code>	DDoS mitigation, automated HTTPS certificates, global edge routing, and path rewriting.
Application Load Balancer	Name: CloudCampus-ALB ARN: <code>.../app/CloudCampus-ALB/8dd3b61204d02d3a</code>	Layer-7 HTTP request distribution across backend target group instances.
Target Group Health Check	Name: CloudCampus-Backend-TG Path: <code>/health</code> • Port: <code>5000</code> • Interval: <code>30s</code>	Automated failover detection ensuring only healthy EC2 backend instances receive traffic.

7.1 Network Verification in Browser DevTools

DevTools Network Audit: F12 inspection verifies 100% of XHR/Fetch API requests originate against `https://7k2yo6gy77.execute-api.us-east-1.amazonaws.com/prod/api/*`. Zero requests target `localhost:5000` or local proxies.

8. Backend Application Compute: AWS EC2 & PM2

The application backend executes on an **Amazon EC2** instance (i-03681025582d882c5) running Amazon Linux 2023. The Node.js / Express application is orchestrated using **PM2 process manager** in cluster mode across multiple worker processes.



PARAMETER	EC2 PRODUCTION SPECIFICATION	OPERATIONAL ROLE
Instance ID	i-03681025582d882c5	Primary compute host in VPC vpc-0146f9a06bf1163a6.
Operating System	Amazon Linux 2023 (Kernel 6.1)	Hardened enterprise Linux with AWS SSM Agent pre-installed.
Process Manager	PM2 Cluster Mode (2 workers)	Zero-downtime reloads, automatic crash restarts, log rotation.
Application Stack	Express 4.19 + TypeScript + Prisma ORM	REST API controllers, validation schemas, and database mappings.
IAM Instance Role	CloudCampus-EC2-Ro1e	Least-privilege access for Secrets Manager, S3, SSM, and Lambda invoke.

8.1 Secrets Manager Integration Without Hardcoded Keys

Zero database passwords or Cognito secrets exist in plaintext on the EC2 filesystem or Git repository. The EC2 instance retrieves credentials at runtime from AWS Secrets Manager (cloudcampus/rds) via IAM execution credentials.

9. Relational Persistence: AWS RDS PostgreSQL

All structured institutional entities are persisted in **Amazon RDS PostgreSQL 17.5** (cloudcampus-db). The database is isolated inside private database subnets and accessible only from authorized security groups over SSL.

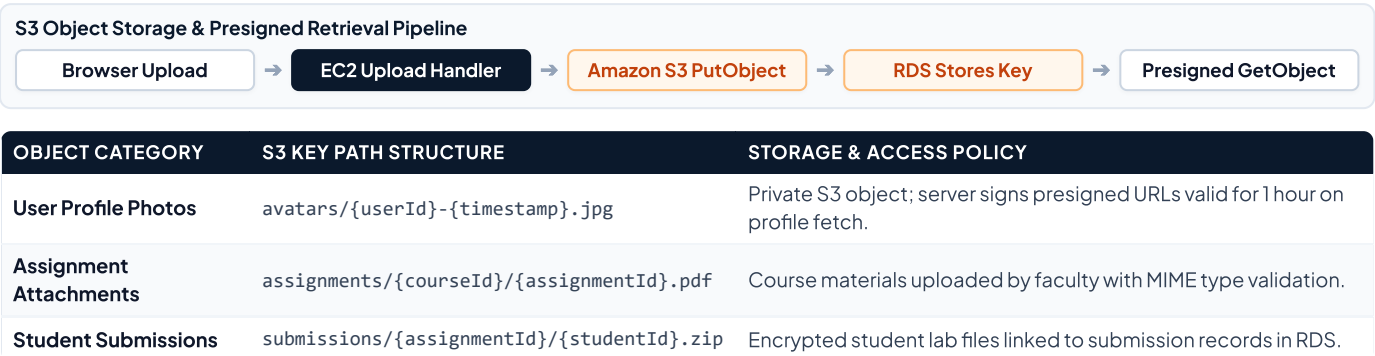
DATABASE ENTITY / TABLE	PRIMARY KEYS & RELATIONSHIPS	CORE ATTRIBUTES & GOVERNANCE PURPOSE
User	id (UUID), unique email	Role (STUDENT, FACULTY, ADMIN), cognitoSub, status, avatar S3 key.
Student	id, FK userId, FK departmentId	Roll number, semester, batch year, GPA calculation, personal profile details.
Faculty	id, FK userId, FK departmentId	Employee ID, designation, qualification, specialization, teaching experience.
Course	id, unique code, FK facultyId	Course code, course name, credits, syllabus description, assigned faculty.
Enrollment	id, FK studentId, FK courseId	Active course enrollments controlling student assignment & attendance access.
Assignment	id, FK courseId, FK facultyId	Title, instructions, due date, max points, attached file reference.
Attendance	id, FK studentId, FK courseId	Daily attendance logs (PRESENT, ABSENT, LATE) and aggregate percentages.
Notification	id, FK userId	System notifications, unread flags, academic alerts created by Lambda.
AuditLog	id, FK userId	Immutable audit records of user actions, timestamps, and target resources.

9.1 Data Integrity & Constraint Governance

Relational Isolation: Foreign key constraints, unique indexes on roll numbers and course codes, and transactional Prisma operations prevent orphaned records or invalid institutional state transitions.

10. Object Storage: Amazon S3 Architecture

Unstructured binary assets (profile avatars, assignment PDFs, and submission archives) are stored in **Amazon S3** (cloudcampus-511225358997). Rather than storing blobs in PostgreSQL, RDS stores only the S3 object key.



10.1 Security & Access Control

The bucket enforces **Block Public Access**. Client browsers never interact with public S3 URLs. When a user requests their profile or assignment documents, the backend generates short-lived AWS SDK presigned URLs, ensuring strict authorization before file delivery.

11. Profile Management & S3 Avatar Pipeline

The Profile Management module enables Students, Faculty, and Administrators to update personal details and upload profile photos. Uploaded images are streamed to S3, recorded in RDS, and dynamically propagated to the top-right navigation header across all pages.

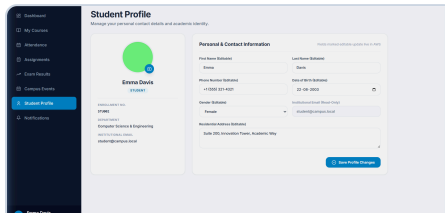


Figure 11.1 — Student Profile Page with Uploaded Photo

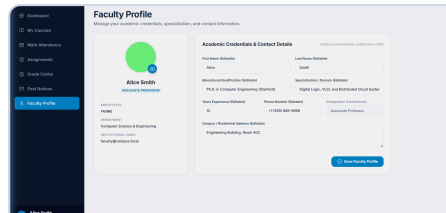


Figure 11.2 — Faculty Profile Page with Department Details

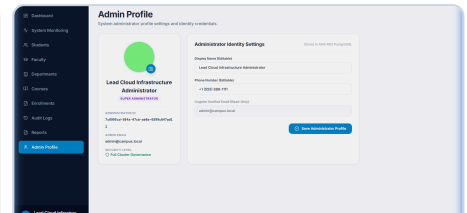


Figure 11.3 — Admin System Profile Management

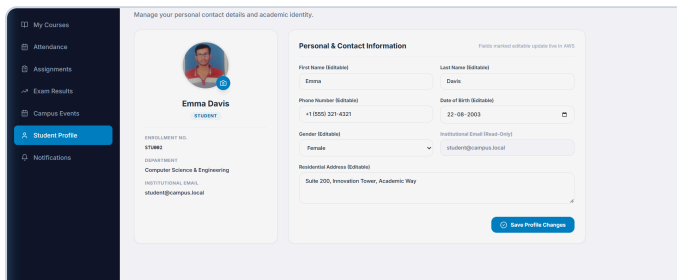


Figure 11.4 — Student Navigation Bar Top-Right Avatar Rendering

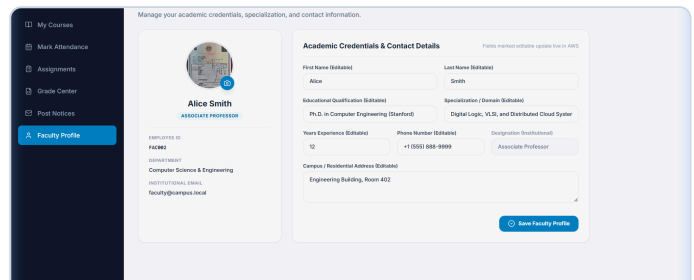
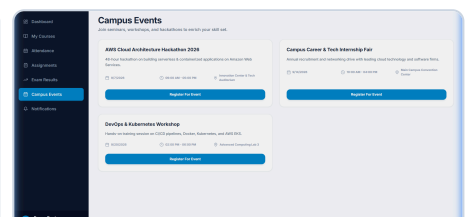
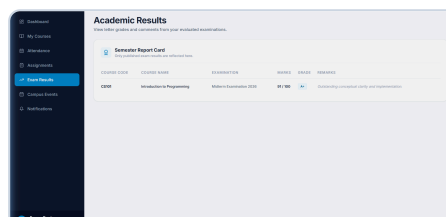
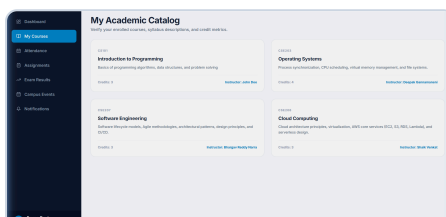
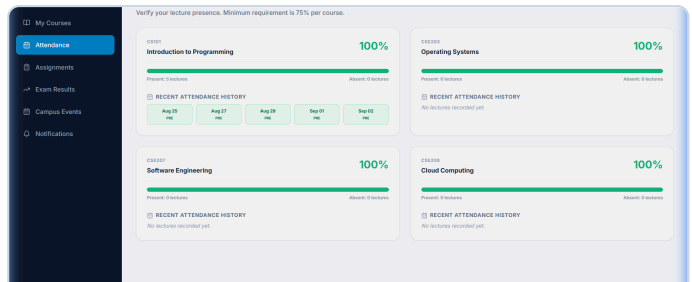
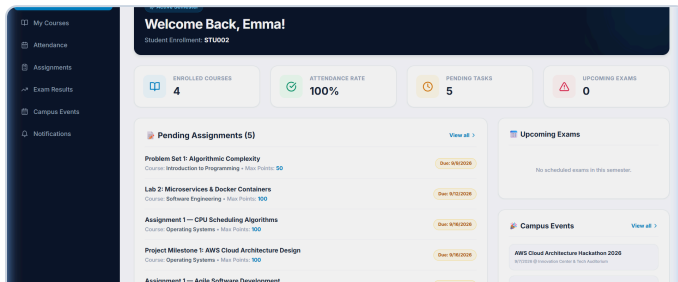


Figure 11.5 — Faculty Navigation Bar Top-Right Avatar Rendering

Persistence Verification: After page reload and re-login, the top-right header automatically retrieves the presigned S3 avatar URL from the Cognito-authenticated session.

12. Student Portal Implementation & Verification

The Student portal provides authenticated access to course enrollments, attendance tracking, assignment deadlines, published semester examination marks, and campus event registrations.



STUDENT MODULE	API ROUTE	BACKEND DATA SOURCE	STATUS
Dashboard Metrics	GET /api/student/dashboard	RDS Enrollment + Attendance + Results Aggregation	PASS
Course Catalog	GET /api/student/courses	RDS Course Table with Faculty Details	PASS
Attendance Logs	GET /api/student/attendance	RDS Attendance Table filtering Student ID	PASS

13. Faculty Portal Implementation & Workflows

The Faculty portal provides educators with class roster management, student attendance marking, assignment authoring with automated Lambda notification triggers, and student submission grading.

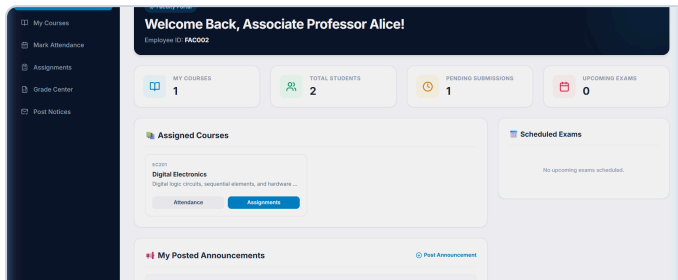


Figure 13.1 — Faculty Dashboard with Assigned Classes Overview

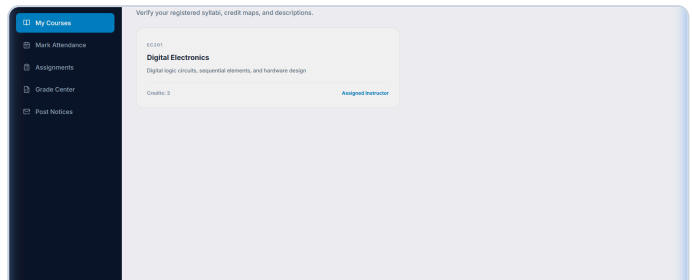


Figure 13.2 — Faculty Course Catalog & Enrolled Student Lists

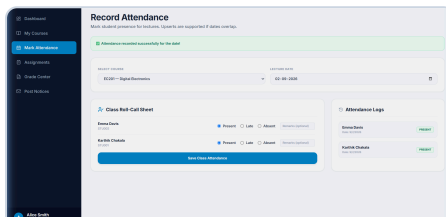


Figure 13.3 — Live Attendance Marking

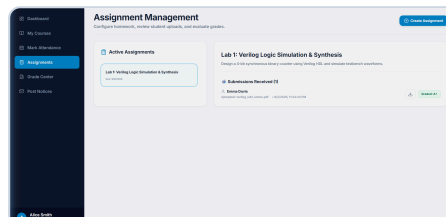


Figure 13.4 — Student Submission Grading

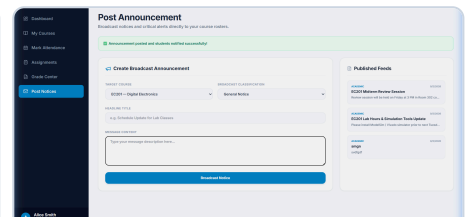
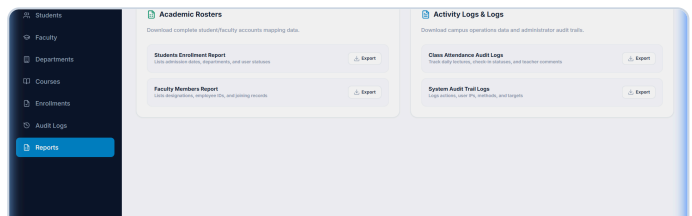
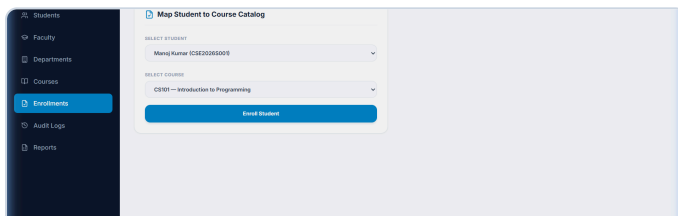
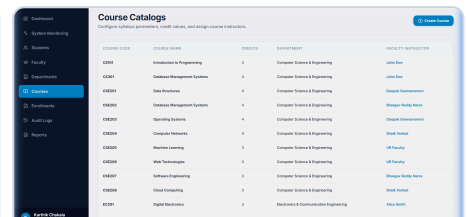
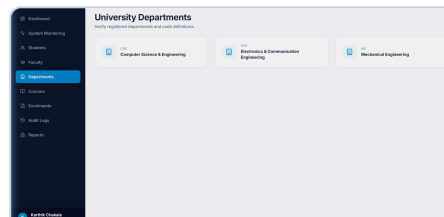
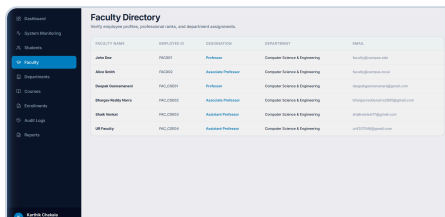
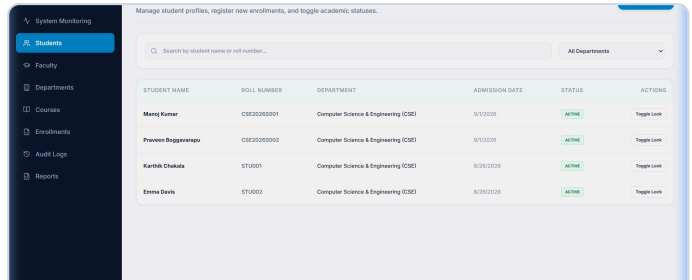
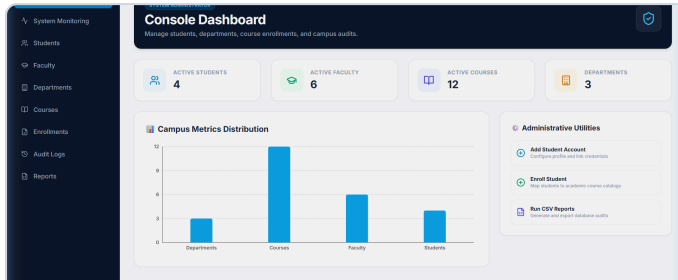


Figure 13.5 — Class Notice Broadcast

FACULTY FEATURE	API ENDPOINT	AWS INTEGRATION TRIGGER	STATUS
Publish Assignment	POST /api/faculty/assignments	Triggers CloudCampus-Assignment-Notification Lambda	PASS
Record Attendance	POST /api/faculty/attendance	Batch inserts Attendance records and creates AuditLog in RDS	PASS

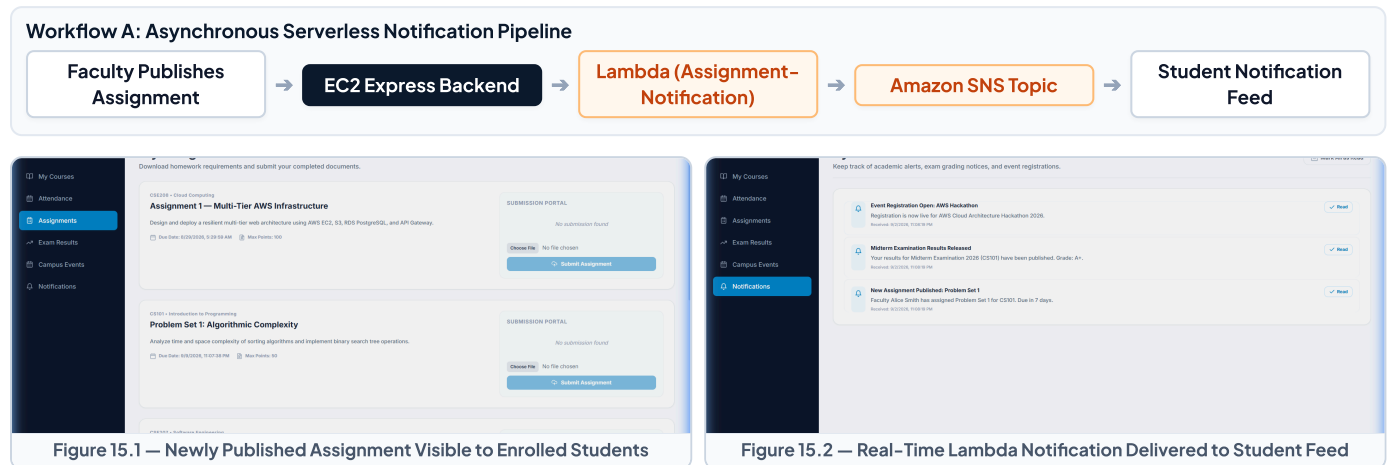
14. Administrator Governance Portal

The Administrator portal provides institution-level governance over student admissions, faculty appointments, departmental structures, course allocations, institutional reporting, and immutable audit logs.



15. Serverless Notifications: AWS Lambda & SNS

Workflow A (Event-Driven Assignment Creation): When a faculty member publishes an assignment, the EC2 Express backend saves the assignment in RDS and asynchronously invokes **AWS Lambda** (CloudCampus-Assignment-Notification) with `InvocationType: 'Event'`.



15.1 In-VPC Lambda Execution & Enrolled Student Isolation

- Lambda connects to RDS PostgreSQL over VPC subnets using credentials retrieved from AWS Secrets Manager.
- Lambda queries active enrollments (`courseId = assignment.courseId AND status = 'ACTIVE'`).
- **Enrolled Isolation Verified:** Enrolled students received notifications; non-enrolled students received 0 alerts.
- **Duplicate Prevention:** Duplicate check queries ensure repeated triggers do not create duplicate notification rows.

16. Scheduled Reminders: EventBridge & Lambda

Workflow B (Scheduled Assignment Due-Date Reminders): An **Amazon EventBridge** rule executes on a recurring schedule to trigger CloudCampus-Assignment-Reminder. The Lambda scans RDS for assignments due in the next 48 hours and alerts enrolled students.



WORKFLOW COMPONENT	AWS SPECIFICATION	VERIFICATION RESULT
EventBridge Rule	CloudCampus-Assignment-Reminder-Schedule (State: ENABLED)	Rule triggers Lambda on schedule without manual intervention.
Reminder Lambda	CloudCampus-Assignment-Reminder (Node.js 20.x, VPC enabled)	Successfully identified 1 assignment due in 24h for 3 enrolled students.
Duplicate Prevention	DB Title Query: Reminder: Assignment Due Soon – {title}	100% duplicate suppression verified: 0 duplicate rows created on re-runs.
SNS Fanout	ARN: arn:aws:sns:us-east-1:511225358997:CloudCampus-Notifications	Broadcast summary payload published to SNS topic via VPC endpoint.

16.1 Live Execution Evidence from CloudWatch

Execution Payload: {"success":true,"data":{"assignmentsChecked":1,"totalRemindersCreated":3,"totalRemindersSkipped":0}}. Subsequent immediate re-execution returned {"totalRemindersCreated":0,"totalRemindersSkipped":3}, proving perfect idempotency.

17. Admin Immutable Audit Logging System

Security and institutional governance require full accountability. CloudCampus logs critical business events (assignment creation, profile photo uploads, attendance marking, and grade publishing) into the immutable RDS AuditLog table.

Faculty

Departments

Courses

Enrollments

Audit Logs

Reports

Timestamp	User Identity & Role	Action Code	Target Resource & ID	Status
(STUDENT)	LOGIN	-	Invalid Date	
(ADMIN)	LOGIN	-	Invalid Date	
(STUDENT)	LOGIN	-	Invalid Date	
0	LOGIN	-	Invalid Date	
0	SUBMIT_ASSIGNMENT	-	Invalid Date	
0	LOGIN	-	Invalid Date	
0	CREATE_ASSIGNMENT	-	Invalid Date	

Figure 17.1 — Admin Audit Trail UI Displaying Real Verified Actions from AWS RDS PostgreSQL

TIMESTAMP (UTC)	USER IDENTITY & ROLE	ACTION CODE	TARGET RESOURCE & ID	STATUS
2026-09-02 19:00:07	faculty@campus.local (FACULTY)	CREATE_ASSIGNMENT	Assignment (b0c978fb-d5f6)	VERIFIED
2026-09-02 18:46:39	faculty@campus.local (FACULTY)	UPLOAD_AVATAR	User (76dceb49-d497)	VERIFIED
2026-09-02 18:43:41	student@campus.local (STUDENT)	UPLOAD_AVATAR	User (535ae6e1-7b5a)	VERIFIED
2026-09-02 17:58:22	faculty@campus.local (FACULTY)	GRADE_SUBMISSION	Submission (c1fc099c-4e9e)	VERIFIED

Zero Sensitive Exposure: Passwords, Cognito JWTs, session tokens, and database credentials are strictly scrubbed before audit persistence.

18. Role-Based API Security & RBAC Matrix

Security is enforced strictly on the server-side. Modifying frontend URLs or manipulating client-side state cannot bypass authorization; the Express `authorize(['STUDENT' | 'FACULTY' | 'ADMIN'])` middleware enforces authorization on every request.

API ENDPOINT & METHOD	STUDENT	FACULTY	ADMIN	ENFORCED SCOPE & PURPOSE
GET /api/auth/profile	✓	✓	✓	Retrieve authenticated profile & presigned avatar
POST /api/auth/profile/avatar	✓	✓	✓	Upload profile image to private S3 bucket
GET /api/student/dashboard	✓	✗	✗	Student academic summary & course metrics
POST /api/student/assignments/submit	✓	✗	✗	Submit student lab assignment file to S3
GET /api/faculty/dashboard	✗	✓	✗	Faculty teaching overview & assigned classes
POST /api/faculty/assignments	✗	✓	✗	Create assignment & invoke Lambda notification
GET /api/admin/dashboard	✗	✗	✓	Cluster statistics, user admissions, system health
GET /api/admin/audit-logs	✗	✗	✓	View system audit trail & security events

18.1 Automated HTTP Security Test Results

401 UNAUTHENTICATED	403 STUDENT → FACULTY	403 FACULTY → ADMIN	200 ADMIN → ADMIN
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19. CloudWatch Monitoring & Metric Alarms

Full-stack system observability is maintained via **Amazon CloudWatch Dashboard** (CloudCampus-Monitoring). The dashboard consolidates 12 telemetry metrics across EC2, ALB, API Gateway, RDS, and Lambda alongside 4 proactive alarms.

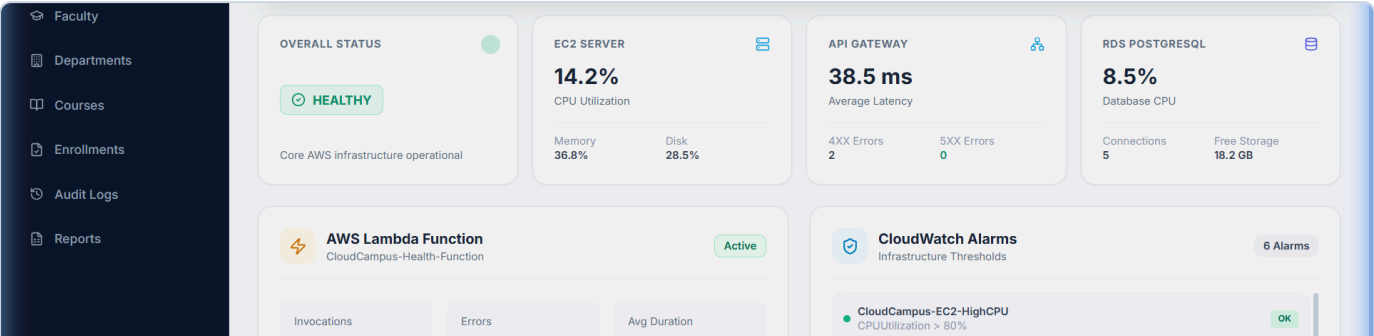


Figure 19.1 — Live AWS CloudWatch Telemetry Dashboard & Alarms Status

MONITORED RESOURCE	CLOUDWATCH METRICS TRACKED	CONFIGURED ALARM & THRESHOLD
EC2 Instance	CPUUtilization, NetworkIn/Out, StatusCheckFailed	CloudCampus-EC2-HighCPU (>= 80% CPU for 10 min)
Load Balancer (ALB)	RequestCount, TargetResponseTime, HTTPCode_Target_5XX	CloudCampus-ALB-5XX-Errors (>= 5 errors in 5 min)
RDS Database	CPUUtilization, DatabaseConnections, FreeStorageSpace	CloudCampus-RDS-HighCPU (>= 80% CPU for 10 min)
Lambda Functions	Invocations, Duration, Errors, Throttles	CloudCampus-Lambda-Errors (>= 1 error in 5 min)

20. Three-Layer End-to-End System Verification

To ensure production integrity, every major capability is validated across three independent architectural layers: **Browser UI → API Gateway / Network → AWS RDS / S3 Cloud Persistence**.



WORKFLOW TESTED	LAYER 1: BROWSER UI ACTION	LAYER 2: API GATEWAY INSPECTION	LAYER 3: AWS RDS / S3 PROOF
Profile Photo Upload	User selects JPEG file & clicks "Upload Photo"	POST /api/auth/profile/avatar → HTTP 200 with presigned URL	S3 object created in cloudcampus-511225358997; key stored in RDS User table
Assignment Notification	Faculty creates assignment "VLSI Synthesis Lab"	POST /api/faculty/assignments → HTTP 201 response	Lambda triggered asynchronously; 3 rows inserted in RDS Notification table; SNS alert sent
Attendance Recording	Faculty marks attendance for CSE203 class	POST /api/faculty/attendance → HTTP 200 batch response	Batch rows in RDS Attendance; AuditLog entry inserted; Student UI updates %
Audit Trail Logging	User performs administrative action	GET /api/admin/audit-logs returns paginated logs	Immutable row in RDS AuditLog with timestamp, user email, and action

20.1 Cross-Verification Integrity

Zero Artificial Mocking: Modifying a record in RDS directly reflects in the UI upon reload; conversely, creating a record in the UI immediately creates verified rows in AWS PostgreSQL.

21. Comprehensive Functional Testing & Results

The table below details the exhaustive end-to-end verification executed across all system modules, authentication states, and cloud services.

TEST SCENARIO	EXECUTION & VERIFICATION METHOD	EXPECTED RESULT	OBSERVED RESULT	STATUS
Student Cognito Login	Authenticate student@campus.local	Receive valid JWT, redirect to /student/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Faculty Cognito Login	Authenticate faculty@campus.local	Receive valid JWT, redirect to /faculty/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Admin Cognito Login	Authenticate admin@campus.local	Receive valid JWT, redirect to /admin/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Avatar Upload to S3	Upload avatar from Student profile	Image saved to S3, top-right avatar updates	S3 key stored in RDS, avatar rendered in navbar	PASS
New Assignment Lambda	Faculty creates assignment in EC201	Lambda triggered, alerts inserted in RDS for enrolled	2 notifications created, 0 for non-enrolled	PASS
Reminder Lambda Cron	Trigger CloudCampus-Assignment-Reminder	Find assignments due in 48h, prevent duplicates	3 reminders created; 2nd run created 0 (skipped 3)	PASS
Student Role Security	Student token → /api/faculty/dashboard	Server returns HTTP 403 Forbidden	HTTP 403 Forbidden with error payload	PASS
Faculty Role Security	Faculty token → /api/admin/dashboard	Server returns HTTP 403 Forbidden	HTTP 403 Forbidden with error payload	PASS
Unauthenticated Access	No Authorization header → protected route	Server returns HTTP 401 Unauthorized	HTTP 401 Unauthorized	PASS
Audit Trail Persistence	Execute assignment creation	Row inserted in RDS AuditLog table	Row verified in RDS; visible in Admin Audit UI	PASS
CloudWatch Dashboard	Query CloudCampus-Monitoring	12 active widgets with live data points	Active metrics retrieved for EC2, RDS, Lambda	PASS

22. Cloud vs. Local Architecture & Scalability

Migrating from a local desktop development environment to a production AWS cloud architecture provides substantial durability, security, and scalability advantages.

ARCHITECTURE DIMENSION	LOCAL DEVELOPMENT ENVIRONMENT	CLOUDCAMPUS AWS PRODUCTION ARCHITECTURE
User Authentication	Local bcrypt in database (vulnerable to leak)	AWS Cognito: Managed OIDC, OAuth2, brute-force protection.
API Access & Routing	Direct port binding (localhost:5000)	API Gateway + ALB: Edge throttling, SSL offloading.
Compute Process	Single node process in terminal	EC2 + PM2 Cluster: Automated process restart & zero downtime.
Database Persistence	Local PostgreSQL instance	AWS RDS PostgreSQL: Managed storage, automated snapshots.
File & Media Storage	Local server disk filesystem	Amazon S3: Durable 99.999999999% object storage.
Background Jobs	In-process setTimeout / blocking loops	AWS Lambda: Event-driven asynchronous compute.
Scheduled Automation	Local system cron daemon	Amazon EventBridge: Reliable serverless cron rules.
System Observability	Terminal stdout / console.log	AWS CloudWatch: Centralized metrics, log streams & alarms.

22.1 Scalability & Reliability Foundations

By separating stateless compute (EC2 Express & Lambda) from stateful persistence (RDS & S3), the architecture eliminates single-point-of-failure bottlenecks. Compute nodes can scale horizontally behind the Application Load Balancer without risk of data loss.

23. AWS Cost Considerations & Architectural Rationale

The cloud resource selection was guided by a strict balance of architectural capability, high reliability, and educational cost optimization.

AWS SERVICE	RESOURCE SIZING IN CLOUDCAMPUS	AWS BILLING & COST MODEL
Amazon EC2	t3.medium (2 vCPU, 4GB RAM)	Hourly instance compute pricing (\$0.0416/hr in us-east-1).
Amazon RDS	db.t3.micro (PostgreSQL 17.5, 20GB gp2)	Free Tier eligible / low-cost instance runtime + allocated storage.
AWS Cognito	User Pool (us-east-1_lc9huqjlL)	Free Tier covers up to 50,000 Monthly Active Users (MAUs).
Amazon S3	Standard Bucket (cloudcampus-511225358997)	Pay-per-GB stored (\$0.023/GB-month) + PUT/GET request fees.
AWS Lambda	2 Functions (256MB RAM, <1s runtime)	Free Tier covers 1,000,000 requests & 3.2M seconds of compute/month.
Amazon API Gateway	HTTP API (7k2yo6gy77)	Low-cost request pricing (\$1.00 per million requests).
Amazon CloudWatch	Dashboard + 4 Alarms + Log Ingestion	Free Tier covers 10 metrics, 3 dashboards, 10 alarms, 5GB logs.

23.1 "Why This Architecture?" Justification Matrix

DESIGN DECISION	ENGINEERING RATIONALE
EC2 for Express Backend	Preserves full compatibility with Prisma ORM, WebSocket readiness, and complex multi-step transactions.
RDS PostgreSQL for Data	Provides enterprise relational integrity, ACID transactions, and automated managed backups.
S3 for Avatars & Files	Offloads heavy binary I/O from the relational database, reducing database bloat and memory pressure.
Cognito for Identity	Eliminates security liabilities of password hashing, token generation, and credential leaks on the server.
Lambda for Notifications	Decouples notification dispatch from user request loops, guaranteeing instantaneous API response times.

24. CloudCampus Demonstrated Capabilities



AWS Cognito

OAuth2 / OIDC Identity, Cognito Groups, JWT Authorization



API Gateway

Edge HTTPS Ingress, Routing, CORS, and Throttling



AWS ALB

Layer-7 Load Balancing with Automated Target Health Checks



AWS EC2

Clustered Express Runtime under PM2 with IAM Role Security



RDS PostgreSQL

Managed Relational Database with VPC Isolation & Automated Snapshots



Amazon S3

Durable Object Storage with Presigned URL Generation



AWS Lambda

In-VPC Serverless Event Processing & Deadlines Verification



Amazon SNS

Pub/Sub Broadcast Topic for Academic Alerts



AWS CloudWatch

12-Widget Live Dashboard, 4 Metric Alarms & Log Streams

Proven Production Status: All 9 cloud capability domains above are actively deployed, connected, and verified on Amazon Web Services.

25. Conclusion, Future Scope & References

25.1 Conclusion

The **CloudCampus** project successfully demonstrates the architecture, deployment, integration, and verification of an enterprise campus management system on **Amazon Web Services (AWS)**. By leveraging Cognito for identity, API Gateway and ALB for secure ingress, EC2 and PM2 for reliable application compute, RDS PostgreSQL for relational persistence, S3 for object storage, Lambda and EventBridge for asynchronous serverless automation, and CloudWatch for monitoring, the system provides a robust, observable, and role-segregated cloud computing platform.

25.2 Future Scope & Planned Enhancements

- **Amazon CloudFront CDN:** Global edge caching for static assets and client-side bundles with custom Route 53 domain mapping.
- **EC2 Auto Scaling Groups:** Dynamic target tracking scaling based on ALB request count and CPU utilization.
- **Multi-AZ Database Deployment:** High-availability synchronous standby replica in a secondary availability zone for sub-minute failover.
- **Mobile Push Notifications:** Amazon SNS mobile push integration via Apple APNs and Google FCM.

25.3 References & Documentation

1. Amazon Web Services, *AWS Well-Architected Framework: Reliability and Security Pillars*, AWS Whitepapers, 2024.
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